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Relationship between topographic heterogeneity and vegetation patterns in a Californian salt marsh

Morzaria-Luna, H.[1]; Callaway J.C.[2]; Sullivan, G.[3] & Zedler, J.B.[4*]

[1]*Department of Botany, University of Wisconsin-Madison, 430 Lincoln Drive, Madison, WI 53706, USA; E-mail morzarialuna@wisc.edu;* [2]*University of San Francisco, 2130 Fulton Street, San Francisco, CA 94117, USA; E-mail callaway@usfca.edu;* [3]*The Wetlands Initiative, 53 West Jackson Boulevard, Chicago, IL 60604, USA; E-mail gsullivan@wetlands-initiative.org.* [4]*Department of Botany and Arboretum, University of Wisconsin-Madison; *Corresponding author; Fax +16082627509; E-mail jbzedler@wisc.edu*

Abstract

Questions: Are species richness and species abundances higher in the presence of tidal creeks? Do species richness and species abundances vary with plot size?

Location: Intertidal plain of Volcano Marsh, Bahia de San Quintin, Mexico.

Methods: We analysed vegetation patterns in large areas (cells) with tidal creeks (+creek) and without (–creek). We surveyed vegetation cover, microtopography, habitat type, and distance to creeks in nested plots of five sizes, 0.1, 0.25, 1, 2.5, and 10 m^2.

Results: Species richness, frequency, cover, and assemblages differed between ±creek cells. Richness tended to be higher in +creek cells, and cover and frequency of individual species differed significantly between ±creek cells. We found consistent patterns in vegetation structure across plot sizes. We encountered 13 species that occurred in 188 unique assemblages. The most common assemblage had six species: *Batis maritima*, *Frankenia salina, Salicornia bigelovii, S. virginica, Salicornia* spec. and *Triglochin concinna*. This assemblage occurred in ±creek cells and at all spatial scales. Of the most common assemblages all but one were composed of multiple species (3-9 species/plot).

Conclusions: The persistence of vegetation patterns across a 100-fold range in spatial scale suggests that similar environmental factors operate broadly to determine species establishment and persistence. Differences in assemblage composition result from variation of frequency and cover of marsh plain species, particularly *Suaeda esteroa* and *Monanthochloe littoralis*. The recommendation for restoration of Californian salt marshes is to target (and plant) multi-species assemblages, not monocultures.

Keywords: Coastal wetland; Sampling scale; Species assemblage; Tidal creek.

Nomenclature: Hickman (1993).

Introduction

Environmental heterogeneity is thought to have a significant influence on the dynamics and structure of ecological communities (Vivian-Smith 1997). In particular, topographic heterogeneity can create a complex mosaic of substrate with varying structure, hydrology, and chemistry (Bledsoe & Shear 2000). In salt marshes, for example, the most obvious features of topographic heterogeneity at a macroscopic scale are the tidal creeks and their accompanying channel banks, levees and ridges. Tidal creeks control tidal flow into the marsh and thus, influence flooding and drainage. These factors can affect vegetation patterns by imposing environmental constraints on plant growth (Mendelssohn et al. 1981). At the microtopographic scale (tens of centimeters), salt marshes also have hummocks, depressions, pools, and shallow finger channels (Adam 1990). Numerous studies of salt marsh vegetation have related plant species distribution with physical gradients associated with tidal flooding and small changes in elevation (Mahall & Roderic 1976; Snow & Vince 1984; Armstrong et al. 1985; Sanderson et al. 2000).

Natural salt marshes with unaltered topography, as occur in Bahia de San Quintin, Baja California, Mexico, serve as reference systems for restoration of highly fragmented and degraded southern California marshes in the southwestern USA (West 2001). This is an arid region, so salt marshes have no constant freshwater input and experience only seasonal storms, with a mean annual rainfall of about 150 mm (Talley et al. 2000). Of Bahia de San Quintin's seven salt marshes, Volcano Marsh lacks major disturbances, such as closure to tidal flushing and freshwater flooding, which are known to alter vegetation (Ibarra-Obando & Poumian-Tapia 1991; Zedler et al. 2001). This marsh has large areas with and without tidal creeks allowing comparisons at multiple sampling scales. This is important because sampling scale might influence the recognition and interpretation of vegetation patterns

(Wiens 1989; Reed et al. 1993; Stohlgren et al. 1997).

Like other marsh plains in the Californian biogeographic region, Volcano Marsh has about a dozen vascular plant species, making it amenable to study the relationship between topographic heterogeneity and species' spatial patterns (Zedler 2001). In addition, the functional roles of these marsh-plain species and their patterns of biomass and nitrogen accumulation have been studied experimentally (Sullivan & Zedler 1999). At Volcano Marsh, plant species distributions have been documented in relation to elevation and tidal creek edges (Zedler et al. 1999). The distributions of three of the eight common species were strongly influenced by elevation, while four varied in relation to both elevation and the presence of tidal creeks (ibid.). Elevation was a useful but incomplete surrogate for environmental conditions and microtopographic heterogeneity was also important.

These results led us to question the spatial extent of the influence of tidal creeks on the structure of the marsh-plain community, including plant species richness, species abundance (cover), and assemblage composition (within-plot species combinations). A clearer understanding of the spatial scale of co-occurrence of plant species in a natural marsh is useful to identify which species combinations are most likely to achieve natural functioning in salt marsh restoration projects. We hypothesized that (1) species richness and species abundances would be higher in the presence of tidal creeks; and that (2) species richness and species abundances would vary with plot size.

Methods

Sampling design

In May 1999, along with researchers from the Pacific Estuarine Research Laboratory (San Diego State University) we surveyed vegetation in Volcano Marsh (Fig. 1; 32°28' N 115°58' W). The marsh is located at the edge of Kenton Hill, and the marsh substrate is volcanic flow overlain by a thin layer of sandy silt, with many volcanic outcrops (Thorsted 1972). Volcano Marsh covers ca. 100 ha; we divided this area into three sections from north to south (Fig. 1). These sections were of similar size, although the area of marsh plain within each section varied. We divided the marsh plain into 50 m × 50 m cells by laying a grid over an aerial photograph. In each section we selected three pairs of cells such that one cell had one or more first or second order creeks (creeks with no or one inflowing tributaries: '+creek'), and one cell had none (–creek). The average distance between ±creek cells (in each pair) was ca. 50 m.

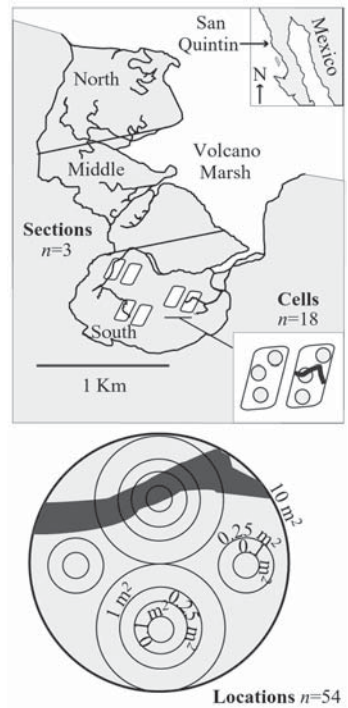

Fig. 1. Sampling design used at Volcano Marsh, Bahia San Quintin, Mexico. The marsh was initially divided into three sections, each of which had three pairs of +creek and –creek cells. Within each cell, we used nested plots of five scales and assessed marsh plain vegetation, as cover by species.

Within each cell we randomly selected three locations; within each we placed a series of circular plots (0.1, 0.25, 1, 2.5, and 10 m^2) in a nested design (Fig. 1). In total, we sampled 540 m^2 across Volcano Marsh; the number of plots varied with sampling scale as shown in Fig. 1. In each plot, we assessed plant cover, classified microtopographic heterogeneity, and habitat type; in cells + creeks we also measured distance to the nearest creek.

We categorized microtopographic heterogeneity as low (< 1 cm difference in elevation within the plot), medium (1-10 cm difference), or high (> 10 cm difference). We classified the types of habitat present within a plot as creek, unvegetated creek, creek slope, unvegetated creek slope, creek edge, level plain, depression, unvegetated depression, and rise. We estimated the percent cover of each plant species and percent bare ground using four cover classes, 1 = < 5%; 2 = 6-25%; 3 = 26-75%; 4 = > 75% (Zedler 2001).

Data analysis

We compared the north, middle, and south sections, cells ±creeks and sampling scales for species richness, species frequency of occurrence, and mean cover class midpoints at 0.1-m^2 to 10-m^2 plots. We used a repeated measures analysis of variance (RMANOVA) to examine nested plots. This procedure implements random effects in the statistical model and incorporates the correlation resulting from adjacent locations (Lin et al. 2000). We had to modify our analysis to account for the varying number of plots at different scales. We used four approaches: (1) eliminated the 0.1-m^2 and 0.25-m^2 plots that did not correspond to larger plots (see Fig. 1) and then applied RMANOVA; (2) averaged species richness of additional 0.1-m^2 and 0.25-m^2 plots within a location and applied RMANOVA; (3) applied an ANOVA Type II; and (4) analysed each scale separately. In each case, we analysed significant differences through the least significant differences test. We included microtopographic heterogeneity and distance to the nearest creek as covariates in these analyses.

We tested for differences in the frequency distribution of species richness between ±creek cells at each plot scale using a modified Kolmogorov-Smirnov (K-S) test (Hamill & Wright 1986); this method identifies exact differences between two distributions. To further test our hypothesis that recorded species richness would vary with the scale sampled, we tabulated the initial number of species found in every 0.1-m^2 plot within the 5-plot nested sets (Fig. 1) and then tabulated the increase from one plot to the next, in the following intervals 0.1-0.25, 0.25-1, 1-2.5, and 2.5-10 m^2. We divided the increase in species number by the increase in area for each interval. The increase in area was 0.15, 0.75, 1.5, and 7.5 m^2, respectively, for the afore-mentioned intervals. We analysed these differences in the increase in mean species number between ±creek cells and between intervals within cells using an ANOVA Type I.

We examined whether species assemblages are influenced by topographic heterogeneity by tabulating assemblages (within-plot species combinations) in cells ± creeks, by scale using the presence/absence matrix. We compared the frequency of each species in the sets of unique assemblages that overlapped among ±creek cells, in assemblages that occurred only in +creek cells and those found only in –creek cells using ANOVA Type I. We tested the significance of the assemblage frequency and the mean number of species per assemblage by comparison with a null-model consisting of 1000 'random' matrices obtained by a Monte Carlo randomization procedure. The matrices were created by randomly allocating species to plots (rows) while maintaining column totals fixed; i.e., the number of species in each scale and the number of times each species occurs are the same in both simulated and observed data (Wilson 1987). We then used the null model to create confidence intervals for the observed data by calculating the number of times the observed frequency of each unique assemblage exceeded its simulated counterpart in the 1000 trials. Significance of these tests was determined by a one-sided test. Thus, to obtain significance at the 5% level, the observed value had to be greater than the random realization at least 950 out of 1000 times. This method of assemblage analysis provides greater detail than the traditional approach of testing for associations between pairs of species (Dale et al. 1991).

To assess microtopography we plotted the frequency distribution of heterogeneity classes and types of habitat present within a plot per ±creek cell. We tested for differences between the distributions of ±creek cells at each spatial scale using the K-S test.

Analyses were carried out using SAS (Anon. 2002) and MATLAB (Anon. 1996) software packages.

Results

Of the 702 plots sampled on the marsh plain of Volcano Marsh, virtually all were vegetated. Only six were unvegetated depressions and creek edges at the 0.1-m^2 and 0.25-m^2 plot scales and only one was unvegetated at the 1-m^2 and 2.5-m^2 scales. In the remaining plots, we found up to 13 plant species (Table 1). We recorded cover separately for each *Salicornia* taxon and examined them separately in the analyses.

Species richness

The cumulative frequency distribution of species richness differed for ±creek cells (K-S test $P = 0.05$) for plot scales 0.1 to 2.5 m^2 (Fig. 2). Mean species richness of ±creek cells consistently exceeded –creek cells (Table 2), although ANOVA did not find significant differences between locations, cells, or marsh sections. Microtopographic heterogeneity was not a significant covariate; for +creek cells distance to a creek was not significant either. Species richness increased significantly with plot size (RMANOVA $F = 95.92$, $P <$ 0001). We also found significant differences (ANOVA

Table 1. Species found on the marsh plain of Volcano Marsh, Bahia de San Quintin, Mexico.

Code	Species
Bm	*Batis maritima*
Cs	*Cuscuta salina*
Fs	*Frankenia salina*
Jc	*Jaumea carnosa*
Lc	*Limonium californicum*
Ml	*Monanthochloe littoralis*
Sb	*Salicornia bigelovii*
Sq	*Salicornia* spec.
Ss	*Salicornia subterminalis*
Sv	*Salicornia virginica*
Sf	*Spartina foliosa*
Se	*Suaeda esteroa*
Tc	*Triglochin concinna*

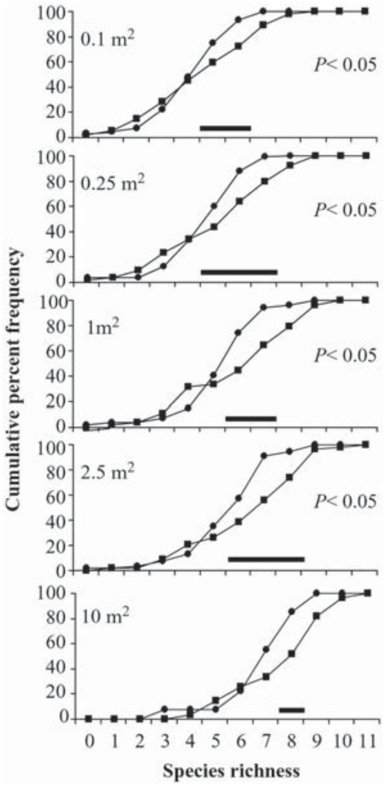

Fig. 2. Cumulative percent frequency of the number of species in plots at Volcano Marsh for five sampling scales ● indicates +creek cellss and ■ –creek cells. Differences in curves assessed with a modified K-S test (Hamill & Wright 1995). Lines above x-axis indicate where significant differences in species richness occur.

+creek cells: F = 25.75, P < 0.001; –creek cells: F = 17.3309, P< 0.001) in the mean increase in species richness as progressively larger plots size intervals were examined; this was true for both cells ± creeks (figure not shown). Differences were largest for the 0.1 - 0.25 m^2 interval even though the greatest change in area was for the 2.5-10 m^2 interval.

Table 2. Mean number of species per plot for all plots (five scales) and for ±creek plots at Volcano Marsh (SE in parentheses). Total sample size for each scale indicated in last row; *n* for ±creek plots = total *n*/2.

Scale	0.1 m^2	0.25 m^2	1 m^2	2.5 m^2	10 m^2
Total	4.65 (0.13)	5.22 (0.13)	5.99 (0.19)	6.36 (0.19)	7.54 (0.23)
+creek plots	4.85 (0.14)	5.48 (0.21)	6.33 (0.04)	6.78 (0.29)	7.92 (0.36)
–creek plots	4.45 (0.1)	4.95 (0.15)	5.63 (0.21)	5.94 (0.16)	7.15 (0.29)
n	216	216	108	108	54

Species abundance and frequency

Five species, *Frankenia salina*, *Jaumea carnosa*, *Limonium californicum*, *Spartina foliosa* and *Salicornia virginica*, had significantly higher cover in +creek cells at one or more spatial scales, while *Salicornia bigelovii*, *Salicornia* spec. and *Suaeda esteroa* had higher cover in –creek cells (App. 1). *S. esteroa* and *F. salina* occurred more frequently in +creek cells at one or more spatial scales, while three species, *L. californicum*, *S. virginica* and *Salicornia* spec. had significantly higher frequency in cells –creek cells (App. 2). In general, patterns of cover and frequency of occurrence agreed for each species from 0.1-m^2 to 10-m^2 plot scales. One exception was *S. esteroa*, which occurred 15 times more frequently in +creek cells than –creek cells but showed on average 70% more cover in –creek cells. The opposite was true for *S. virginica*, which was less frequent in +creek cells than –creek cells, but its mean cover was 113% higher in +creek cells, at the 0.25-, 1-, and 2.5-m^2 scales. *S. virginica* was both more frequent and more abundant in +creek cells at the 0.1-m^2 scale. There were no significant differences in frequency or cover for individual species between marsh sections or locations.

Species assemblages

We found 188 unique within-plot species assemblages among the five spatial scales (Fig. 3); +creek cells had 120 assemblages, with a mean of 3.3 (0.5 SE) assemblages in each location (plot scales combined), while 96 assemblages occurred in –creek cells, with a mean of 3.02 (0.45 SE) assemblages in each location. In both ±creek cells 28 assemblages occurred, compared to 92 found only in +creek cells and 68 assemblages that occurred uniquely in –creek cells.

Assemblages ranged from one to 11 species; the only species found alone was *S. virginica*. The most common

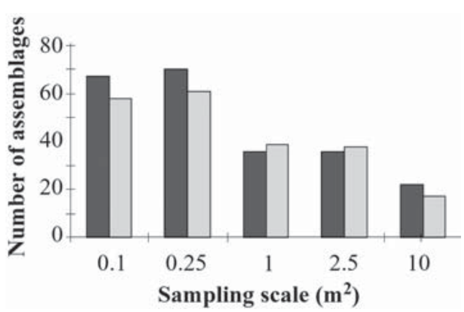

Fig. 3. Assemblage (within plot species combinations) frequency and specific composition in Volcano Marsh. The graph shows abundance of assemblages in each sampling scale. Dark bars indicates +creek cells and clear bars –creek cells.

assemblage had six species, and this assemblage was present in 13 locations across the marsh plain; 52 % of assemblages were found in three or more locations. Species frequencies differed significantly among the three sets of assemblages, those that overlapped in ±creek cells, those found only in –creek cells, and only in +creek cells (ANOVA: $F = 9.35$, $P < 0.001$).

The distribution of assemblage frequencies observed at Volcano Marsh differed significantly from that of the null model. Up to 60% of the 28 assemblages shared by ±creek cells were more common than in the simulated null model at least 95% of the time; 22 assemblages occurred in 40% of the 702 plots sampled for all scales combined; 11 of these are listed in Table 3. Six of the 22 assemblages are unique to +creek cells, four to –creek cells, and 11 overlap in ±creek cells. Of these widespread assemblages, four prevailed across all five plot-scales (Table 3). The remaining assemblages were rare.

Microtopographic heterogeneity and habitat types

The microtopography was more heterogeneous in the presence of tidal creeks. The frequency histograms of elevation difference within plots (Fig. 4) show that, on average 25.5% of plots surveyed in +creek cells had high heterogeneity, while none of the plots surveyed in –creek cells had this level of microtopographic complexity. On average, 87.6% of plots in –creek cells had low microtopographic heterogeneity versus 60.7% in +creek cells. The plots sampled in +creek cells differed significantly in the distribution of microtopographic heterogeneity from those in –creek cells at the 0.1, 1, 2.5 and 10-m² scales (K-S test: $P = 0.05$).

The most common type of habitat was level plain (525 occurrences among all plot scales), followed by creek edge (148), depression (61) and creek slope (59).

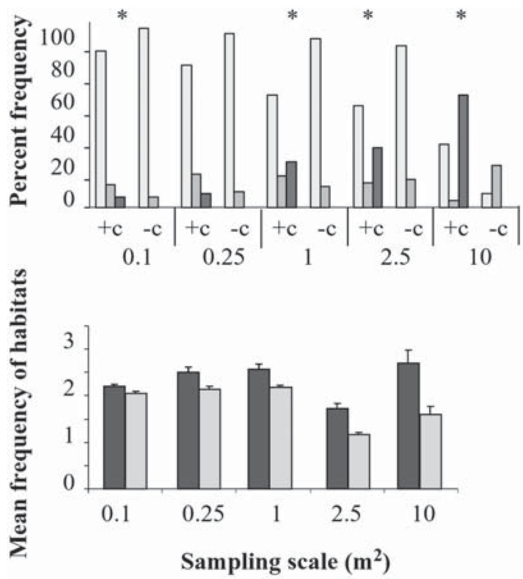

Fig. 4. Microtopography and frequency of habitat classes in sampling plots. Top graph is frequency of microtopography (habitat heterogeneity) classes found in plots of five sampling scales. Clear bars are low heterogeneity (< 1 cm difference in elevation within the plot), medium bars are medium (1-10 cm difference), and dark bars are high (> 10 cm difference). Significant differences in the distribution of types of habitat in ±creek cells are indicated * = $P < 0.05$ (Kolmogorov-Smirnov test). Bottom graph is mean frequency of habitat types present within plots. Dark bars: +creek cells; clear bars: –creek cells.

The remaining classification categories accounted for 13% of the types of habitat encountered. The distribution of types of habitat present in +creek cells was significantly different from the distribution in –creek cells at all sampling scales (K-S test: $P = 0.05$; Fig. 4).

Table 3. Composition and frequency of occurrence for assemblages found in five or more locations across the marsh plain of Volcano Marsh. Frequency is indicated separately for ±creek cells (sizes pooled). The scale at which the assemblages were not present is indicated.

No. of species	Assemblage composition	Absent at scale (m²)	Frequency locations			Total
	±creek cells		+creek cells	–creek cells	Total	
6	Bm-Fs-Sb-Sq-Sv-Tc		14	23	37	13
7	Bm-Fs-Lc-Sb-Sq-Sv-Tc		9	21	30	12
5	Bm-Sb-Sq-Sv-Tc	10	5	23	28	11
9	Bm-Fs-Jc-Lc-Sb-Se-Sq-Sv-Tc		16	0	16	7
1	Sv	10	8	1	9	5
3	Bm-Fs-Sv		12	12	24	5
4	Bm-Sb-Sq-Tc	1, 2.5, 10	3	5	8	5
5	Bm-Lc-Sb-Sq-Tc	10	1	8	9	5
6	Bm-Fs-Lc-Sb-Sq-Tc	10	3	3	6	5
7	Bm-Fs-Ml-Sb-Sq-Sv-Tc	0.1, 0.25, 10	0	8	8	5
8	Bm-Fs-Lc-Ml-Sb-Sq-Sv-Tc	0.1, 0.25, 1	0	6	6	5

Discussion

Vegetation patterns differ in ±creek cells

We found that species richness, frequency, cover and assemblages differ between ±creek cells and that these effects extend beyond the 1-m edge effects documented by others (Hopkins & Parker 1984; Zedler et al. 1999) and the creek order relationship reported for northern California marshes (Sanderson et al. 2000). Our results show that +creek cells tend to have higher species richness than –creek cells (Fig. 2 and Table 2). The difference in species richness found at the 0.25 m^2 scale, namely, 5.48 (0.21 SE) species in +creek cells and 4.45 (0.01 SE) species in –creek cells, is higher than the channel edge effect previously described by Zedler et al. (1999). In that study, the lowest elevation class had an average of 3.9 species ≤ 1 m from a creek, compared to 3.4 species > 1 m from creek. In turn, the difference found here among ±creek cells is lower than the variation reported in northern California. Hopkins & Parker (1984) found an increase of four species ≥ 5 m from a creek, and Sanderson et al. (2000) documented a gain of 1.6 species ≤ 10 m from natural creeks. The relationship between species richness and distance to tidal creeks has been related to gradients of tidal flooding and drainage (Hopkins & Parker 1984; Zedler et al. 1999; Smith et al. 2002). The fact that distance to a creek was not significant as a covariate in our data might have been due to fewer sampling plots and a lower chance of detecting creek edge effects.

As expected, several species differed significantly in both cover and frequency of occurrence between cells ± creeks (Apps. 1 and 2). While perennials mostly grew best in +creek cells, the annuals *S. bigelovii* and *Salicornia* spec. grew better in –creek cells where canopy cover of perennials tends to be low. Lower canopy cover should favor germination of seeds by allowing greater light penetration. Lower cover of perennials could be explained by poorer drainage in –creek cells. Where drainage is impeded, soils become waterlogged, microbial activity reduces oxygen (Baldwin & Mendelssohn 1998) and leads to the production of sulfides and reduced iron and manganese (Ingold & Havill 1984). As a result, redox potential decreases and inhibits plant growth (Armstrong et al. 1985).

Marsh species have differential tolerances to aeration and flooding, dependent on internal ventilating efficiencies, capacity for rhizosphere oxidation, response to altered nutrient regimes and ability to metabolize in anoxic conditions (Armstrong et al. 1985). *F. salina* had both significantly higher cover and occurred more frequently in +creek cells. This highly productive species extends its range into the wetland-upland transition (Sullivan & Zedler 1999) where soils are only wetted by rainfall. *S. virginica* had significantly higher cover in +creek cells, but occurred more frequently in cells – creeks. This species has the broadest ecological tolerance, with high frequency from low to high intertidal elevations and ability to dominate salt marshes following loss of tidal influence (Zedler et al. 2001).

Although we did not measure environmental variables in this study, others have related patterns of spatial distribution and dominance in Pacific Coast salt marshes to environmental gradients of inundation (Mahall & Roderic 1976), elevation (Snow & Vince 1984) and salinity (Callaway et al. 1990). Distribution patterns of salt marsh species have also been attributed to tidal dispersal of seeds and physical disturbance. The factors that influence seedling establishment (high soil salinity, waterlogging, and iron and sulfide toxins) are ameliorated along creek edges (Brown & Bledsoe 1996), and seedling numbers decline with distance from creeks (Hopkins & Parker 1984). Creeks also provide a corridor for the transport of wrack (wood and dead plant material) onto the marsh, and its deposition creates open patches for recruitment (Fischer et al. 2000). These factors explain why *S. esteroa,* a short-lived species that relies on seedling recruitment, is common along creek edges at Volcano Marsh.

Vegetation patterns were consistent across spatial scales, with most variation at the fine scale

If different ecological processes are dominant at different spatial scales, plot size can significantly affect the interpretation of vegetation patterns (Reed et al. 1993). This has been shown in studies of species-rich systems (grasslands with + 300 species) or using larger plot sizes (0.1-100 ha) (Stohlgren et al. 1997). In contrast, we found consistent differences in cover and frequency of occurrence between ±creek cells across the 0.1 to 10-m^2 plot scales (Apps. 1 and 2). Furthermore, five assemblages that occurred in > 5 locations prevailed across sampling scales (Table 3). The small species pool of the Californian marsh plain (eight common species) might result in scale independence, with similar patterns and processes at the five scales we considered (Wiens 1989).

Most of the spatial variability in Volcano Marsh occurred at the smallest plot scales. Species richness increased most as plot size moved from 0.1-0.25 m^2 and least for the 2.5-10 m^2 interval. If, as concluded by Reed et al. (1993), composition of large plots reflects the broad, coarse variations in the physical environment, while small plots pick up plant-to-plant interactions, then low variability among our larger plots implies consistent cause-effect relationships across most of the scales we assessed (Underwood & Petraitis 1993). Biotic interactions would appear to have their greatest effect at the 0.1-m^2 scale.

Assemblages reflect the frequency and cover of marsh plain species

The most common assemblage we found was composed of six species (Table 3). Each of these species contributes unique attributes to the assemblage (Sullivan & Zedler 1999). *B. maritima*, a trailing succulent with a stem and a few side branches, produces sparse cover low to the ground so there is bare ground underneath for the germination and establishment of the annuals *S. bigelovii* and *Salicornia* spec. (Zedler 1977). *S. virginica* is a productive upright succulent with tall shoots; thus, it has a more complex canopy structure with higher potential for light competition (Sullivan & Zedler unpubl.) and the capacity to trap wrack and sediment, adding nutrients (Adam 1990). *T. concinna* and *F. salina* have high root biomass and a deep rhizosphere (Sullivan & Zedler 1999); these species might be better able to capture limiting nutrients and accumulate more resources below-ground (Cody 1986). *T. concinna* has a high root : shoot ratio (Sullivan & Zedler 1999) and as a slow growing species with high tissue nitrogen, it might accumulate biomass over time and strongly influence nutrient dynamics through seasonal uptake and release (Berendse 1994). Each of the species contributes functions that are impaired when diversity declines (Zedler et al. 2001).

As shown in Table 3, many differences in assemblage composition between ±creek cells result from the presence or absence of *S. esteroa* and *M. littoralis*. These species have different habitat preferences. *M. littoralis* is a native perennial grass usually found at higher elevation, 0.74-2.3 m NGVD, on the high marsh, while *S. esteroa* is a short-lived perennial subshrub found at 0.62 to 1.35 NGVD (Sullivan & Noe 2001). Frequency of occurrence and cover of *S. esteroa* were significantly higher in +creek cells than in –creek cells for all sampling scales (Apps. 1 and 2). This result is in accord with Sullivan's (2001) finding of high mortality of *S. esteroa* seedlings where water ponded in a Mission Bay, CA, restoration site. Zedler et al. (2003) also found low survival rates for *S. esteroa* seedlings transplanted to a Tijuana Estuary restoration site with high soil salinity.

Findings are relevant to salt marsh restoration

We agree with Zedler et al. (1999) and Sanderson et al. (2000) that tidal creeks should be incorporated into restoration designs. We also recommend that when creating or restoring salt marshes, managers plant assemblages or combination of species, not monoculture patches. The recurring assemblages found on the marsh plain of Volcano Marsh (Table 3) can be used to develop planting mixes for ±creek areas.

The most common assemblage was composed of *B. maritima*, *F. salina*, *S. bigelovii*, *Salicornia* spec., *S. virginica*, and *T. concinna* (Table 3). Because *Salicornia* spec. has only been documented in Volcano Marsh, this species is not recommended for restoration projects elsewhere. The remaining five species could be planted as a cluster within 0.1-m^2 plots. If the restoration site is near an existing salt marsh that has abundant *S. virginica*, it will probably not be necessary to plant this species, as it disperses and establishes easily (Lindig-Cisneros & Zedler 2002). The planting of multi-species assemblages would contribute to the natural marsh functions of seedling recruitment (Lindig-Cisneros & Zedler 2002), development of complex canopies (Keer & Zedler 2002) and biomass and nitrogen accumulation (Callaway et al. 2003). The planting of different assemblages in various parts of the restoration site could constitute experiments that test hypotheses about which assemblages are best suited to each type of habitat or microsite.

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